CS513: Theory & Practice of Data Cleaning
Group Project Instructions (Summer 2026)

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The goal of the group project is to conduct an end-to-end data cleaning project, using the various tools and techniques that we have covered throughout the course. In addition to the main tools that we used in class (i.e., RegEx, OpenRefine, Datalog/Logica, SQL, and Python), you are welcome to use other tools. For example, you may want to use any of the research prototypes mentioned (e.g., YesWorkflow or the OpenRefine companion tools such as or2yw), or commercial tools (e.g., Tableau). Or maybe you find a new way to use an LLM (e.g., ChatGPT). In all these cases your report will have to include sufficient documentation about how you used these tools.

Project Phases. The project is organized into two phases, each with its own set of deliverables. All milestone dates (*Teams Announced*, *Teams Locked*, *Phase-I Due*, and *Phase-II Due*) are published in the **Key Dates** posting on Coursera and Campuswire.

Project Phase-I

During this phase your team needs to ...

1. **Describe your chosen dataset** D. Choose one of the available datasets from the **Box folder**: (1) the New York Public Library (NYPL) *"What's on the Menu?"* historic menus data, and (2) the *Chicago Food Inspection* (CFI) data.[1] To describe the data, you should provide a **conceptual model** (ER diagram) that depicts the entity and relationship types, or an **ontology** that illustrates all classes and their relationships. Or you can provide a **database schema** that illustrates and explains the structure and contents of the dataset. You should also add a short **narrative**, i.e., one or more paragraphs in English to describe the origin of the data and any relevant metadata (e.g., a *temporal* or *spatial extent*).

2. **Develop three use cases.** By *use case* we mean a written scenario describing a hypothetical data analysis. (If helpful, you can think of these as a queries or questions asked of the dataset, see **Additional Information** below).

   (a) **Target (main) use case**: U_1 for D such that data cleaning is *necessary* and *sufficient* to support the data analysis use case. Thus, after performing data cleaning, your cleaned data D' is *fit-for-purpose* (i.e., for U_1).

   (b) **"Zero data cleaning" use case**: U_0 should be a use case that requires "zero data cleaning", i.e., D is "good enough as it is".

   (c) **"Never enough" use case**: U_2 is a use case for which the given dataset D is "never (good) enough", i.e., no amount of data cleaning or wrangling will make D suitable for U_2 (even though at first sight one might think so).

[1] See the Coursera and/or Campuswire posting for further details and come to the Office Hours if necessary.

(d) **Note**: The purpose of the corner cases U_0 (data cleaning is **not necessary**) and U_2 (data cleaning is **not sufficient**) is to reinforce the concept that **data cleaning should be done with a purpose in mind**, i.e., a use case such as your main use case U_1, where data cleaning really makes a difference.

3. **List obvious data quality problems** (i.e., which are easy to spot during Phase-I). In order for your dataset D and main use case U_1 to match, data cleaning must be necessary and sufficient to implement U_1. You need to **support this claim** by **documenting** data quality problems that your inspection of D has revealed and that need to be addressed before U_1 can be tackled.

4. **Devise an initial plan** that outlines how you intend to clean the dataset in Phase-II. A typical plan for the overall project will include the following **steps**: S_1: description of dataset D and matching use case U_1; S_2: profiling of D to identify the quality problems P that need to be addressed to support U_1; S_3: performing the data cleaning process using one or more tools to address the problems P (here you should describe which tools you are planning to use, e.g., OpenRefine; Python; etc.) S_4: checking that your new dataset D' is an improved version of D, e.g., by documenting that certain problems P are now absent and that U_1 is now supported; S_5: documenting the types and amount of changes that have been executed on D to obtain D'.

   You should also include a tentative **assignment of tasks** to team members (who does what)!

Additional Information

Regarding (2): How do you specify data analysis **use cases**? Generally speaking, you can simply explain the use case in a short paragraph. You might also want to be more specific and phrase use cases as **questions**: *What is it that we want to know from (or about) the data?*

In particular, a use case may be a set of database **queries** $Q_1, \ldots, Q_n$ against the dataset D (e.g., how many farmers markets offer bakery goods in addition to vegetables and fruits?) On the other hand, use cases may also be more general, e.g., you could state that you'd like to develop a web application that serves a particular purpose.

The advantage of specifying a use case U as one or more queries Q_U is that you can be very precise about when data cleaning is necessary and sufficient for U: if running Q_U on the original ("dirty") data D would result in an answer $A = Q_U(D)$ that is incorrect and/or misleading, then data cleaning is **necessary**. Conversely, data cleaning is **sufficient** if the answer $A = Q_U(D')$ on the cleaned dataset D' is correct (and not misleading).

In (3) above, how do you document data quality problems? One simple way is to include (copy-pasted) snippets of "dirty data" in your Phase-I report (you can also use screenshots for illustration) and then explain what the problem is in narrative form.

How do you describe your **plan** in (4)? A short list of your planned steps $S_1, \ldots, S_5$ will do during Phase-I. In Phase-II, you should also include a **workflow diagram** for the actual data cleaning steps that you performed (e.g., with YesWorkflow or any other diagramming tool). Of course your Phase-I **plan** and your **actual** Phase-II **workflow** might be different.

Project Phase-II

During this phase you will **execute** the plans you've come up with in Phase-I (possibly adjusting course, based on what you find when actually working with the data ...)

What to Submit

Phase-I:

- **A single PDF file with your Phase-I report** (in narrative form) with all elements from the list above.

Phase-II:

- **A single PDF file with your Phase-II report**. This report should include:
  - A description of the **actual data cleaning workflow** W that was performed, and a **comparison** with the original Phase-I plan: e.g., were you able to execute the steps as planned, and if not, what did you have to change and why?
  - A **narrative** that ties all steps together and explains the motivation (use case U_1), the rationale for the design of the overall workflow W and the tools used.
  - **Documentation** that data quality was improved, e.g., through running "before queries" $Q_U(D)$ and "after queries" $Q_U(D')$ on D (original) and D' (cleaned), respectively.
  - A summary of the **data changes** ΔD resulting from the overall workflow $W\colon D \rightsquigarrow D'$.
  - A summary of **findings**, **problems encountered**, and **lessons learned**, including **possible next steps** (e.g., how would you implement the main use case U_1).
- **Supplementary Materials**. In addition to the project report, you need to provide the following supplementary materials (as a single ZIP file):
  1. **Workflow Model**: For the **overall** workflow model W, when using YesWorkflow, provide the text file that has the YW annotations (e.g., `Workflow.yw`), and the generated Graphviz (dot) file (e.g., `Workflow.gv`). For other diagramming tools, provide a source file (e.g., PPTX, ...) and a PDF file (`Workflow.pdf`).
  2. **OpenRefine Operation History**: If you used OpenRefine, then include a copy of the *operation history* (copy-paste it into a `json` file named `OpenRefineHistory.json`).
     - If you also want to visualize the OpenRefine history, you can use the `or2yw` tool.[2]
  3. **Other History**: If you are using an **alternative tool** (instead of, or in addition to OpenRefine), please provide an analogous file (`OtherToolHistory.json`) and other provenance information **if available for that tool**: e.g., include Python (or R) scripts, Jupyter notebook files, etc.
  4. **Queries**: A copy of the queries written in SQL or Datalog to profile the dataset and check the integrity constraints (copy-paste them into a **text file** named `queries.txt`).
  5. **Original ("dirty") and Cleaned Datasets**: Please **do not** provide the datasets in the ZIP file. Rather, upload the raw and cleaned datasets in a Box folder and **share the link** in a plain text file (`DataLinks.txt`).

[2] https://github.com/idaks/OR2YWTool

Project Phase-I Report Outline and Checklist

We strongly encourage you to follow the Phase-I report outline below, as it aligns well with the checklist and grading rubric. The title / header of your Phase-I report should list (i) the Team-ID of your group[3] and (ii) for each group member the name and Illinois email address.

Report Outline / Checklist

The project instructions (separate document above!) describe what exactly you should do as part of Phase-I of the project. The following outline should be followed for your Phase-I report:

1. **Description of Dataset** (25 points)
   - (a) Here you will provide an ER diagram, an ontology, or a detailed database schema (10 points), and
   - (b) a narrative description of the dataset covering structure and content (15 points)
2. **Use Cases** (30 points)
   - (a) Target (Main) use case U_1: data cleaning is necessary and sufficient (20 points)
   - (b) "Zero data cleaning" use case U_0: data cleaning is not necessary (5 points)
   - (c) "Never enough" use case U_2: data cleaning is not sufficient (5 points)
3. **Data Quality Problems** (30 points)
   - (a) List obvious data quality problems with evidence (examples and/or screenshots) (20 points)
   - (b) Explain why / how data cleaning is necessary to support the main use case U_1 (10 points)
4. **Initial Plan for Phase-II** (15 points)
   - (a) Below is a possible plan, listing typical data cleaning workflow steps. In your Plan for Phase-II, fill in additional details for the project steps as needed. In particular, include who of your team members will be responsible for which steps, and list the timeline that you are setting yourselves!
     - S_1: Review (and update if necessary) your use case description and dataset description
     - S_2: Profile D to identify DQ problems: How do you plan to do it? What tools are you going to use?
     - S_3: Perform DC "proper": How are you going to do it? What tools do you plan to use? Who does what?
     - S_4: Data quality checking: is D' really "cleaner" than D?
       - Develop test examples / demos
     - S_5: Document and quantify change (e.g. columns and cells changed, IC violations detected: before vs. after, etc.)

[3] In addition you can give yourself a (cute) team name for ease of identification ;-)

Project Phase-II Report Outline and Checklist

We strongly encourage you to follow the Phase-II report outline below, as it aligns well with the checklist and grading rubric. The title / header of your Phase-II report should list (i) the Team-ID of your group and (ii) for each group member the name and Illinois email address.

Report Outline / Checklist

The project instructions (separate document above!) describe what exactly you should do as part of Phase-II of the project. The following outline should be followed for your Phase-II report:

1. **Description of Data Cleaning Performed**
    - Identify and describe all (high-level) data cleaning steps you have performed. (20 points)
    - For each high-level data cleaning step you have performed, explain its rationale. Was the step really required to support use case U_1? Explain. If not, explain why those steps were still useful. (20 points)
2. **Document data quality changes**
    - Quantify the results of your efforts, e.g., by providing a summary table of changes: Which columns changed? How many cells (per column) have changed, etc.? (10 points)
    - Demonstrate that data quality has been improved, e.g., by devising IC-violation reports (answers to denial constraints) and showing the difference between number of IC violations reported before and after cleaning. (10 points)
3. **Create a workflow model**
    - A visual representation of your overall (or "outer") workflow W_1, e.g., using a tool such as YesWorkflow. At a minimum, you should identify key inputs, outputs, and steps of the workflow, along with dependencies between these. Key phases and steps of your data cleaning project may include, e.g., data profiling, data loading, data cleaning, IC violation checks, etc. Explain the design of W_1 and why you've chosen the tools that you have in your overall workflow. (10 points)
    - A detailed (possibly visual) representation of your "inner" data cleaning workflow W_2 (e.g., if you've used OpenRefine, you can use the OR2YW tool). (10 points)
4. **Conclusions & Summary** (10 points)
    - Please provide a concise summary and conclusions of your project, including lessons learned.
    - Reflect on how work was completed. You should summarize the contributions of each team member here (for teams with ≥ 2 members).
5. **Submission of supplementary materials in a single ZIP file** (10 points)
    - Workflow Model
    - Operation history
        - OpenRefine Recipe
        - Other scripts, provenance files
    - Queries
    - Original ("dirty") and Cleaned datasets
        - Please provide an accessible Box folder link in a plain text file: `DataLinks.txt`

Group Project Submission Rules and Policies

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Submission Procedure

- Each team can submit the Phase-I report, the Phase-II report, and the supplementary materials **ONLY ONCE**.
- The reason is that (unlike for auto-graded assignments) these are manually graded. Once review and grading has begun, we cannot start over. Therefore, please only submit the **FINAL** version of your deliverables!

Team Composition

- Immediately after your team is announced (see *Teams Announced* in the **Key Dates** posting), reach out to all your team members to confirm that your schedules and time zones work out and that you are committed to working together on the project!
- If for some reason the survey-based team assignment does not work out, please reach out to TAs via Campuswire with the group information: Create a post with the category "Group Project" and available to "Instructors & TAs". Submit any such request before the **Teams Locked** date (see the **Key Dates** posting on Coursera/Campuswire). Include a detailed justification why the team does not work. After **Teams Locked**, all teams are final!